QORVO PROFESSIONAL TRAINING

Design Verification - DV26

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Requirement & Functional Specification

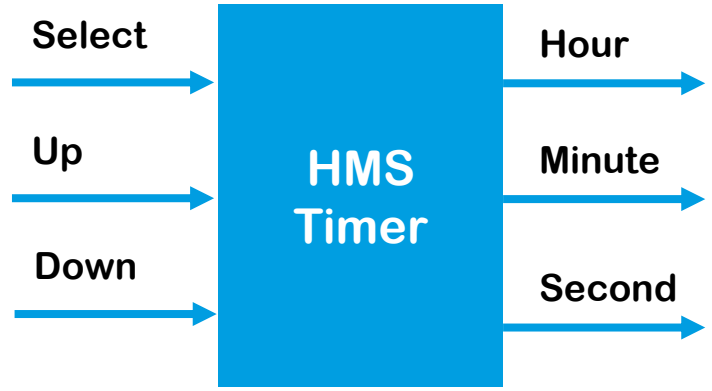
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Hour-Minute-Second Timer

Requirement & Functional Specification

❖ Design a simple Hour-Minute-Second Timer

- ❑ Output the real time value via Hour, Minute, Second output signals
- ❑ Time value can be adjusted by using Select, Up, Down input signals
  - *Select: select to adjust hour, minute, second value*
  - *Up: Increase time value*
  - *Down: Decrease time value*
- ❑ A qualified low-to-high transition of Select, Up, or Down generates a one-clock internal trigger
- ❑ 1Mhz input clock and asynchronous active low reset

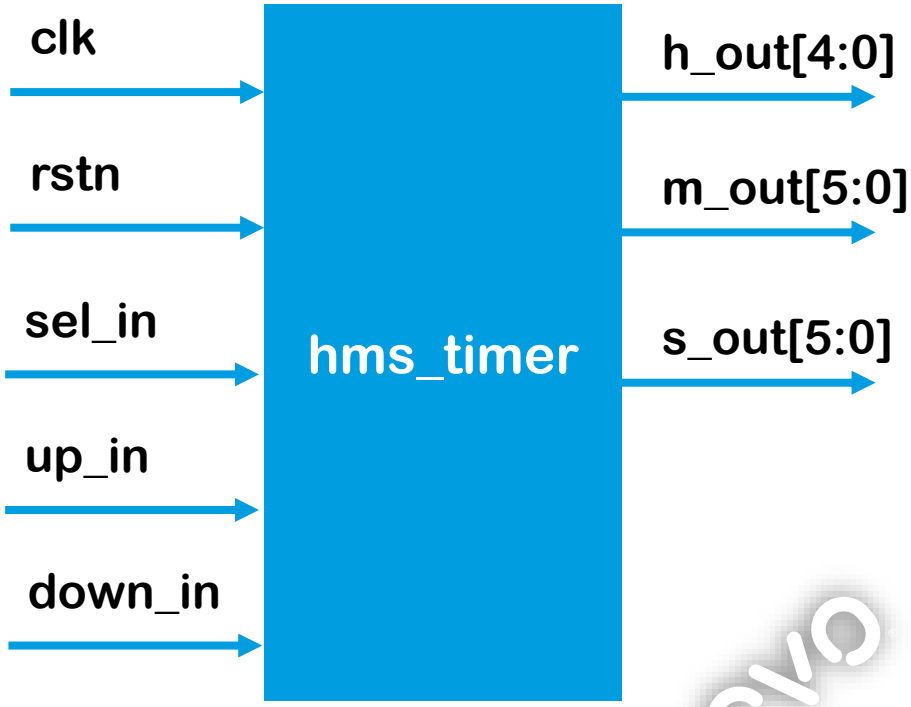


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Hour-Minute-Second Timer

I/O

Port Name	I/O	Width	Description
clk	In	1	Clock signal
rstn	In	1	Reset signal (Asynchronous active low)
sel_in	In	1	Raw asynchronous Select control input
up_in	In	1	Raw asynchronous increment control input
down_in	In	1	Raw asynchronous decrement control input
h_out	Out	5	Output hour value
m_out	Out	6	Output minute value
s_out	Out	6	Output second value



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Hour-Minute-Second Timer

Time Format & Reset

❖ 24-hour format

- ❑ Hour = 00 to 23
- ❑ Minute = 00 to 59
- ❑ Second = 00 to 59
- ❑ Rollover: 23:59:59 → 00:00:00
- ❑ Hour, Minute, and Second outputs use unsigned binary encoding

❖ Power cycle

- ❑ No power retention
- ❑ After power-up, rstn shall be asserted before normal operation. The timer shall start at 00:00:00 in NORMAL mode
- ❑ All internal divider, synchronizer, trigger-history, and trigger state shall reset to 0

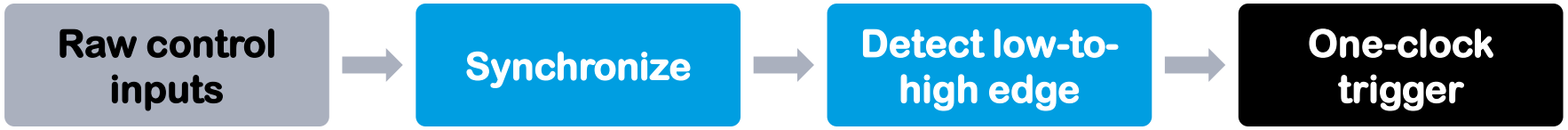
At reset (rstn = 0)	Value
h_out	0
m_out	0
s_out	0
FSM state	NORMAL

Current time	After one-second tick
12:34:58	12:34:59
12:34:59	12:35:00
12:59:59	13:00:00
23:59:59	00:00:00

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Hour-Minute-Second Timer

Control Input Handling

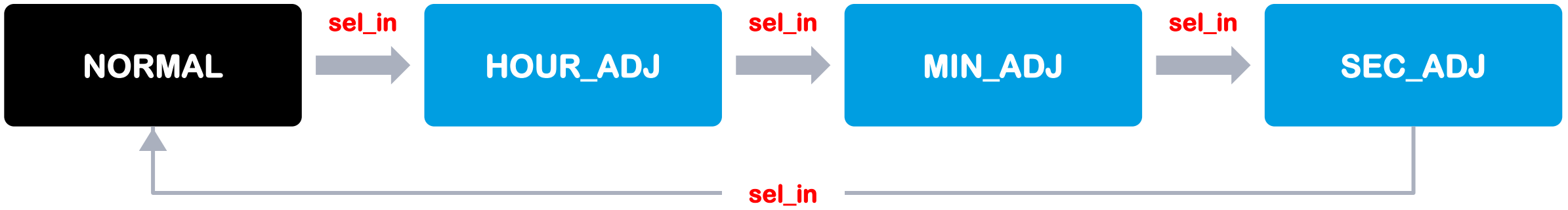


- The sel_in, up_in, and down_in inputs may be asynchronous to clk
  - Each control input shall pass through a two-stage synchronizer before being used by the HMS functional logic
    - *The external source shall hold each asserted and de-asserted control input level stable for at least two complete clk cycles*
  - Note: To simplify the design, deglitching and debouncing are not supported in this version
- Trigger = one action
  - sel_in: One sel trigger pulse causes exactly one FSM state transition
  - up_in / down_in: One up trigger or down trigger pulse changes the selected field by exactly one unit
  - A high-to-low transition generates no functional trigger
- Input held high
  - No repeated transitions or adjustments; auto-repeat is not supported
- Clocking
  - FSM and counters update on a rising edge of clk
  - One-Second Time Base - Input clock frequency: 1 MHz
    - *One second consists of exactly 1,000,000 rising edges of clk*

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Hour-Minute-Second Timer

FSM



- Transitions
  - One sel trigger pulse causes one state transition
  - Reset state is NORMAL; no adjustment-mode timeout
- Counting while adjusting
  - Normal time counting is paused in all adjustment states
  - Subsecond divider keeps its value and resumes from it in NORMAL

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Hour-Minute-Second Timer

Manual Adjustment

State	Selected field	Wrap on up_in	Wrap on down_in
HOUR_ADJ	Hour	23 → 00	00 → 23
MIN_ADJ	Minute	59 → 00	00 → 59
SEC_ADJ	Second	59 → 00	00 → 59

❖ Step

- ❑ One qualified edge of up_in or down_in changes the selected field by exactly one unit
- ❑ In NORMAL mode, up trigger and down trigger are ignored

❖ Wrap

- ❑ Only the selected field wraps; no carry or borrow into adjacent fields
- ❑ Natural counting still performs normal carry

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Hour-Minute-Second Timer

Simultaneous Control Events

Ignore events
All control events dropped

Hold state
FSM state unchanged

Hold time
Hour, Minute, Second unchanged

Pause counting
No time count that cycle

- Trigger
  - More than one trigger among sel_in, up_in and down_in on the same rising edge of clk
- Reset priority
  - rstn low overrides everything: Hour, Minute, Second = 0, state = NORMAL

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Thank You